

Practice Questions for Game Tree

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1 Questions

1. According to the lecture, what is the primary purpose of searching over game trees in the context of AI game playing?
 - a. To memorize all possible game states
 - b. To identify the optimal sequence of moves leading to a win
 - c. To randomly select moves to confuse the opponent
 - d. To create a visual representation of the game
 - e. To calculate the probability of each move
2. According to the lecture, what is a key limitation of using brute-force search for complex games like Chess or Go?
 - a. It requires too much memory
 - b. It is too slow to be practical
 - c. It is not able to learn from past games
 - d. It is not able to represent the game state
 - e. It is not able to reason about the opponent's moves
3. According to the lecture, what is the role of leaf nodes in a game tree?
 - a. They represent the starting position of the game
 - b. They represent intermediate game states
 - c. They represent the possible moves a player can make
 - d. They represent the final outcomes of the game with associated scores
 - e. They represent the opponent's possible moves
4. According to the lecture, what is the main purpose of planning sequences of actions (look ahead) in game playing algorithms?
 - a. To randomly select moves
 - b. To predict the opponent's next move
 - c. To evaluate the current game state

- d. To anticipate future game states and their potential outcomes
 - e. To memorize all possible game states
5. a. According to the lecture, what is the primary purpose of searching over game trees in the context of AI game playing?
- A. To identify the optimal sequence of moves leading to a win
 - B. To represent the rules of the game
 - C. To store all possible game states
 - D. To visualize the game board
 - E. To calculate the probability of winning
- b. According to the lecture, what is a key limitation of using brute-force search for complex games like Chess or Go?
- A. It requires too much memory
 - B. It is too slow to be practical
 - C. It cannot handle stochastic games
 - D. It cannot represent the game state
 - E. It is not able to learn from past games
- c. According to the lecture, what is the role of leaf nodes in a game tree?
- A. They represent the starting position of the game
 - B. They represent intermediate game states
 - C. They represent the possible moves
 - D. They represent the final outcomes of the game
 - E. They represent the opponent's moves
- d. According to the lecture, what is the main purpose of planning sequences of actions (look ahead) in game playing algorithms?
- A. To make random moves
 - B. To react to the opponent's moves
 - C. To predict the opponent's moves
 - D. To evaluate the current game state
 - E. To anticipate future game states and their outcomes
6. According to the lecture, what is the primary purpose of searching over game trees in the context of AI game playing?
- a. To identify the optimal sequence of moves leading to a win
 - b. To randomly explore all possible game states
 - c. To evaluate the current board configuration
 - d. To determine the opponent's next move
 - e. To create a visual representation of the game

7. According to the lecture, what is the role of leaf nodes in a game tree?
 - a. They represent the starting position of the game
 - b. They represent intermediate game states
 - c. They represent the possible moves a player can make
 - d. They represent the final outcomes of the game with associated scores
 - e. They represent the opponent's possible moves
8. According to the lecture, what is a key limitation of using brute-force search for complex games like Chess or Go?
 - a. It requires too much memory
 - b. It is too slow to find a solution
 - c. The search space is too large to explore exhaustively
 - d. It is not able to represent the game state
 - e. It is not able to learn from past games
9. According to the lecture, what is the main purpose of planning sequences of actions (look ahead) in game playing algorithms?
 - a. To randomly select the next move
 - b. To evaluate the current board state
 - c. To anticipate future game states and their potential outcomes
 - d. To memorize all possible game sequences
 - e. To determine the opponent's current move
10.
 - a. According to the lecture, what is the primary purpose of searching over game trees in the context of AI game playing?
 - A. To identify the optimal sequence of moves leading to a win
 - B. To represent the current state of the game
 - C. To define the set of legal moves
 - D. To reason about the opponent's possible actions
 - E. To choose between actions
 - b. In a game tree, what do the leaf nodes typically represent?
 - A. The initial state of the game
 - B. Intermediate game states
 - C. Possible moves
 - D. Outcomes of the game with associated scores
 - E. The current player's turn

- c. According to the lecture, why is brute-force search alone insufficient for complex games like Chess or Go?
 - A. Because the rules of these games are too complex
 - B. Because the search space is too large
 - C. Because computers are not fast enough
 - D. Because these games require human intuition
 - E. Because these games have too many players
- d. What is the main purpose of planning sequences of actions (look ahead) in game playing algorithms?
 - A. To represent the current state of the game
 - B. To identify the set of legal moves
 - C. To anticipate future game states and outcomes
 - D. To reason about the opponent's current move
 - E. To choose between actions

2 Answers

1. According to the lecture, what is the primary purpose of searching over game trees in the context of AI game playing?
 - a. Answer A is incorrect because memorizing all possible game states is not feasible for complex games due to the size of the search space.
 - b. Answer B is correct because the primary goal of searching over game trees is to find the best sequence of moves that will lead to a win.
 - c. Answer C is incorrect because randomly selecting moves is not a strategic approach to game playing.
 - d. Answer D is incorrect because while game trees can be visualized, the primary purpose is not visualization.
 - e. Answer E is incorrect because while probabilities can be used, the primary purpose is to find the optimal sequence of moves.
2. According to the lecture, what is a key limitation of using brute-force search for complex games like Chess or Go?
 - a. Answer A is incorrect because while memory is a factor, the primary limitation is the time it takes to search the space.
 - b. Answer B is correct because the lecture states that the search space for complex games is too large, making brute-force search computationally infeasible.
 - c. Answer C is incorrect because brute-force search does not involve learning, but rather exhaustively searching all possibilities.
 - d. Answer D is incorrect because brute-force search can represent the game state, but it cannot search the entire space in a reasonable time.
 - e. Answer E is incorrect because brute-force search does not reason about the opponent's moves, it simply searches all possible moves.
3. According to the lecture, what is the role of leaf nodes in a game tree?
 - a. Answer A is incorrect because the starting position is represented by the root node.
 - b. Answer B is incorrect because intermediate game states are represented by the internal nodes of the tree.
 - c. Answer C is incorrect because possible moves are represented by the branches of the tree.
 - d. Answer D is correct because leaf nodes in a game tree represent the final outcomes of the game, such as a win, loss, or draw, and are associated with a score.
 - e. Answer E is incorrect because the opponent's moves are represented by the branches of the tree, not the leaf nodes.
4. According to the lecture, what is the main purpose of planning sequences of actions (look ahead) in game playing algorithms?
 - a. Answer A is incorrect because planning sequences of actions is not about randomly selecting moves, but rather making strategic choices.

- b. Answer B is incorrect because while predicting the opponent's moves is important, the main purpose is to anticipate future game states.
 - c. Answer C is incorrect because evaluating the current game state is a separate step, not the main purpose of planning sequences of actions.
 - d. Answer D is correct because the purpose of planning sequences of actions is to look ahead and anticipate the future states of the game and their potential outcomes.
 - e. Answer E is incorrect because memorizing all possible game states is not feasible for complex games.
5. a. According to the lecture, what is the primary purpose of searching over game trees in the context of AI game playing?
- A. Answer A is correct because the primary goal of searching over game trees is to find the best sequence of moves that will lead to a win.
 - B. Answer B is incorrect because the rules of the game are used to generate the game tree, not the purpose of searching it.
 - C. Answer C is incorrect because while game trees represent game states, the purpose of searching is not to store them all, but to find the best path.
 - D. Answer D is incorrect because game trees are an abstract representation of the game, not a visualization of the board.
 - E. Answer E is incorrect because while the probability of winning might be a factor, the primary purpose is to find the optimal sequence of moves.
- b. According to the lecture, what is a key limitation of using brute-force search for complex games like Chess or Go?
- A. Answer A is incorrect because while memory is a factor, the primary limitation is the time it takes to search the tree.
 - B. Answer B is correct because the search space for complex games is so large that brute-force search becomes computationally infeasible due to the time it would take.
 - C. Answer C is incorrect because brute-force search can be applied to stochastic games, but it is still limited by the size of the search space.
 - D. Answer D is incorrect because brute-force search can represent the game state, but it is not efficient.
 - E. Answer E is incorrect because brute-force search does not learn from past games, but that is not the primary limitation.
- c. According to the lecture, what is the role of leaf nodes in a game tree?
- A. Answer A is incorrect because the starting position is represented by the root node.
 - B. Answer B is incorrect because intermediate game states are represented by the internal nodes of the tree.
 - C. Answer C is incorrect because possible moves are represented by the branches of the tree.
 - D. Answer D is correct because leaf nodes in a game tree represent the final outcomes of the game, such as a win, loss, or draw.
 - E. Answer E is incorrect because the opponent's moves are represented by the branches of the tree, not the leaf nodes.

- d. According to the lecture, what is the main purpose of planning sequences of actions (look ahead) in game playing algorithms?
- A. Answer A is incorrect because planning sequences of actions is not about making random moves, but about making strategic moves.
 - B. Answer B is incorrect because while reacting to the opponent's moves is important, planning sequences of actions is about anticipating future states, not just reacting.
 - C. Answer C is incorrect because while predicting the opponent's moves is a part of planning, the main purpose is to anticipate future game states and their outcomes.
 - D. Answer D is incorrect because evaluating the current game state is a part of the process, but the main purpose is to anticipate future states.
 - E. Answer E is correct because the main purpose of planning sequences of actions is to look ahead and anticipate future game states and their outcomes, allowing the algorithm to make more informed decisions.
6. According to the lecture, what is the primary purpose of searching over game trees in the context of AI game playing?
- a. Answer A is correct because the primary goal of searching over game trees is to find the best sequence of moves that will lead to a win.
 - b. Answer B is incorrect because while exploration is part of the process, the goal is not to randomly explore all states, but to find the optimal path.
 - c. Answer C is incorrect because while evaluating the board is important, it is not the primary purpose of searching the game tree.
 - d. Answer D is incorrect because while reasoning about the opponent's moves is important, the primary purpose is to find the best moves for the AI player.
 - e. Answer E is incorrect because while game trees can be visualized, the primary purpose is not to create a visual representation.
7. According to the lecture, what is the role of leaf nodes in a game tree?
- a. Answer A is incorrect because the starting position is represented by the root node, not the leaf nodes.
 - b. Answer B is incorrect because intermediate game states are represented by the internal nodes of the tree, not the leaf nodes.
 - c. Answer C is incorrect because possible moves are represented by the branches of the tree, not the leaf nodes.
 - d. Answer D is correct because leaf nodes in a game tree represent the final outcomes of the game, such as a win, loss, or draw, and each outcome has an associated score.
 - e. Answer E is incorrect because the opponent's moves are represented by the branches of the tree, not the leaf nodes.
8. According to the lecture, what is a key limitation of using brute-force search for complex games like Chess or Go?
- a. Answer A is incorrect because while memory is a factor, the primary limitation is the size of the search space.

- b. Answer B is incorrect because while speed is a factor, the primary limitation is the size of the search space.
 - c. Answer C is correct because the search space for complex games is so large that it is computationally infeasible to explore all possible game states using brute-force search.
 - d. Answer D is incorrect because brute-force search can represent the game state, but it cannot explore all possible states in a reasonable time.
 - e. Answer E is incorrect because brute-force search does not learn from past games, but this is not the primary limitation.
9. According to the lecture, what is the main purpose of planning sequences of actions (look ahead) in game playing algorithms?
- a. Answer A is incorrect because planning sequences of actions is not about randomly selecting the next move, but about making informed decisions.
 - b. Answer B is incorrect because while evaluating the current board state is important, planning sequences of actions is about looking beyond the current state.
 - c. Answer C is correct because planning sequences of actions, or looking ahead, allows the algorithm to anticipate future game states and their potential outcomes, enabling it to make more strategic decisions.
 - d. Answer D is incorrect because memorizing all possible game sequences is not feasible for complex games, and planning sequences of actions is about exploring potential future states.
 - e. Answer E is incorrect because while reasoning about the opponent's moves is important, planning sequences of actions is about anticipating future game states.
10. a. According to the lecture, what is the primary purpose of searching over game trees in the context of AI game playing?
- A. Answer A is correct because the primary goal of searching over game trees is to find the best path through the possible moves to achieve a win.
 - B. Answer B is incorrect because representing the current state is a part of the game playing algorithm, but not the purpose of searching over game trees.
 - C. Answer C is incorrect because defining legal moves is a part of the game rules, not the purpose of searching over game trees.
 - D. Answer D is incorrect because reasoning about the opponent's actions is a part of the game playing algorithm, but not the purpose of searching over game trees.
 - E. Answer E is incorrect because choosing between actions is a part of the game playing algorithm, but not the purpose of searching over game trees.
- b. In a game tree, what do the leaf nodes typically represent?
- A. Answer A is incorrect because the initial state is represented by the root node, not the leaf nodes.
 - B. Answer B is incorrect because intermediate states are represented by the internal nodes of the tree, not the leaf nodes.
 - C. Answer C is incorrect because possible moves are represented by the branches of the tree, not the leaf nodes.
 - D. Answer D is correct because leaf nodes in a game tree represent the final outcomes of the game, each with a score indicating the result (win, lose, draw).

- E. Answer E is incorrect because the current player's turn is not represented by the leaf nodes.
- c. According to the lecture, why is brute-force search alone insufficient for complex games like Chess or Go?
- A. Answer A is incorrect because while the rules can be complex, the primary issue is the size of the search space.
 - B. Answer B is correct because the lecture emphasizes that the search space for complex games is astronomically large, making brute-force search computationally infeasible.
 - C. Answer C is incorrect because while computer speed is a factor, the search space is so large that even with faster computers, brute-force search is not feasible.
 - D. Answer D is incorrect because while human intuition is valuable, the lecture focuses on the computational limitations of brute-force search.
 - E. Answer E is incorrect because the number of players is not the reason brute-force search is insufficient.
- d. What is the main purpose of planning sequences of actions (look ahead) in game playing algorithms?
- A. Answer A is incorrect because representing the current state is a separate component of the algorithm.
 - B. Answer B is incorrect because identifying legal moves is a part of the game rules, not the purpose of planning sequences of actions.
 - C. Answer C is correct because planning sequences of actions (look ahead) is about anticipating future game states and outcomes to make informed decisions.
 - D. Answer D is incorrect because reasoning about the opponent's current move is a part of the algorithm, but not the purpose of planning sequences of actions.
 - E. Answer E is incorrect because choosing between actions is the final step, not the purpose of planning sequences of actions.